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Mingyu Sung

AI Research Engineer — Efficient GenAI Inference (Quantization · Caching · Acceleration)

Nota AI, Platform Team (Quantizer) · South Korea

AI Research Engineer at Nota AI, working on efficient inference for LLMs and diffusion models — quantization, KV-cache compression, cache-based acceleration and CUDA kernel optimization. Ph.D. in Artificial Intelligence, Kyungpook National University (2026). Core contributor to the Nunchaku CUDA acceleration engine (ICLR 2025 Spotlight), with 16 merged PRs across Nunchaku, Hugging Face Transformers, ByteDance DreamO and other open-source projects.

EXPERIENCE

AI Research Engineer — Platform Team, Quantizer Part 2026.02 – Present

Nota AI

- Develop Nota's LLM/VLM quantization platform: post-training quantization (GPTQ, AWQ, SmoothQuant, rotation-based), KV-cache quantization, and low-bit export (NVFP4, GGUF).
- Extend model support and evaluation across LLM, VLM, MoE, embedding and video-diffusion models.
- Optimized K-EXAONE 236B (MoE) for FuriosaAI's data-center NPU with LG AI Research: ~71% smaller model at ~99.2% accuracy retention.

EDUCATION

Ph.D., Artificial Intelligence 2021.08 – 2026.02

Kyungpook National University (KNU) · Thesis: Fast and Trustworthy Super-Resolution with VLM-Guided Diffusion Model

M.S., Computer Science 2019.09 – 2021.08

Kyungpook National University (KNU) · Thesis: Probabilistic Classification Method of Spiking Neural Network Based on Multi-Labeling of Neurons

B.S., Computer Science 2013.03 – 2019.08

Kyungpook National University (KNU)

OPEN SOURCE

Nunchaku — Core Contributor 2025 – 2026-02

nunchux-ai/nunchaku · 9 merged PRs · CUDA acceleration engine for 4-bit neural networks (ICLR 2025 Spotlight, 3.9k+ stars)

- Cache optimizations — V2 FBCaching (#621), double FB cache, TeaCache batch processing (#601) — for 2–5x additional speedup
- IP-Adapter (XLabs flux-ip-adapter-v2) support (#418)
- Fixes: LoRA key mismatch (#557), ControlNet (#360, #452), Sana (#380), offload segfault (#440)
- Forward-pass tensor-op cleanup (#491)

ComfyUI-nunchaku — Contributor 2025

nunchux-ai/ComfyUI-nunchaku · 2 merged PRs · ComfyUI integration for the Nunchaku engine

- IP-Adapter support for ComfyUI (#305)
- Cache-mechanism file separation (#474)

DreamO — Contributor 2025

bytedance/DreamO · 1 merged PRs · ByteDance's unified image-customization framework

- Dynamic FBCache / DoubleFBCache support for Nunchaku engine integration (#104)

Transformers — Contributor 2026

huggingface/transformers · 3 merged PRs · Hugging Face Transformers library

- Fixed float16 overflow in Gemma4 vision pooler (#46277)
- Fixed Gemma `sliding_window` being halved on every config save/reload, a silent accuracy regression for EmbeddingGemma (#47940)
- Fixed `_init_weights` reading a quantized child's `weight`, which broke loading CLIP-family quantized checkpoints (#47921)

LoopQuant — Contributor 2026

Kishon-zzx/LoopQuant · 1 merged PRs · Post-training quantization for looped language models (Ouro)

- Fixed the LCTC teacher load resetting the ILPA per-loop activation-bit allocation, which ran the W4A6 configs at A4 in every loop; Ouro-2.6B W4A6 WikiText-2 perplexity 8.10 → 7.85 (#1)

PUBLICATIONS

[J1] A Bi-Objective Framework for Scene Text Super-Resolution: Balancing Legibility and Fidelity
M. Sung, S. Ham, K. Kim, I.-M. Kim, S. Yun, J.-M. Kang · Transactions on Machine Learning Research (TMLR) · 2026

[J2] H2-Cache: A Novel Hierarchical Dual-Stage Cache for High-Performance Acceleration of Generative Diffusion Models
M. Sung, I.-M. Kim, S. Yun, J.-M. Kang · IEEE Open Journal of the Computer Society · 2025

[J3] A Novel VLM-Guided Diffusion Model for Remote Sensing Image Super-Resolution
M. Sung, M.-G. Gong, S.-J. Ham, I.-M. Kim, S. Yun, J.-M. Kang · IEEE Geoscience and Remote Sensing Letters · 2025

[J4] DeCo-MeSC: Deep Compression-Based Memory-Constrained Split Computing Framework for Cooperative Inference of Neural Network
M. Sung, V. Palakonda, I.-M. Kim, S. Yun, J.-M. Kang · IEEE Transactions on Vehicular Technology · 2025

[J5] Generative Diffusion Model-Based Deep Learning Framework for Remaining Useful Life Prediction
S. Ha, **M. Sung**, F. Saeed, S. Yun, I.-M. Kim, J.-M. Kang · IEEE Internet of Things Journal · 2025

[J6] Entropy-based sampling for efficient training of deep learning on CNC machining dataset
M. Sung, C. Park, S. Ha, M. Ha, H. Lee, J. Kim, J.-M. Kang · Electronics Letters · 2024

[J7] Probabilistic Classification Method of Spiking Neural Network Based on Multi-Labeling of Neurons
M. Sung, J. Kim, J.-M. Kang · Mathematics · 2023

[J8] Training and Inference using Approximate Floating-Point Arithmetic for Energy Efficient Spiking Neural Network Processors
M. Kwak, J. Lee, H. Seo, **M. Sung**, Y. Kim · ICEIC · 2021

[J9] Training spiking neural networks with an adaptive leaky integrate-and-fire neuron
M. Sung, Y. Kim · IEEE ICCE-Asia · 2020

PREPRINTS

[P1] RECAP-KV: Single-Prefill Approximation of Reconstruction Importance for Reusable KV Cache Eviction
OpenReview · 2026

[P2] Memory- and Latency-Constrained Inference of Large Language Models via Adaptive Split Computing
arXiv:2511.04002 · 2025

[P3] No Pose Estimation? No Problem: Pose-Agnostic and Instance-Aware Test-Time Adaptation for Monocular Depth Estimation
arXiv:2511.05055 · 2025

[P4] Why Should the Server Do It All?: A Scalable, Versatile, and Model-Agnostic Framework for Server-Light DNN Inference over Massively Distributed Clients via Training-Free Intermediate Feature Compression
arXiv:2511.11608 · 2025

[P5] Range Asymmetric Numeral Systems-Based Lightweight Intermediate Feature Compression for Split Computing of Deep Neural Networks
arXiv:2511.11664 · 2025

SELECTED PROJECTS

K-EXAONE 236B MoE Optimization (Nota AI × LG AI Research × FuriosaAI) ↗ — Quantization Engineer · 2026
Optimized the 236B-parameter MoE model for FuriosaAI's data-center NPU via targeted precision analysis of degradation-prone sections — ~71% size reduction with ~99.2% accuracy retention (GPQA 79.80 / IFBench 68.98 / AIME25 88.57).

LLM Inference Optimization (Samsung Challenge) — Lead Developer · 2024
Optimized pipeline for Phi-3 using Torch-TensorRT conversion and memory-aware dynamic batching to accelerate LLM inference.

Advanced CNC Machine Tool Diagnosis & Prognosis (KERI) — Lead Developer · 2023
Deep learning models for tool wear classification, anomaly detection, and RUL prognosis.

AI-based Intelligent CCTV (ABB Project) — AI Optimization Engineer · 2023
Optimized an LSTR behavior-detection model for edge devices via Knowledge Distillation (TransKD) and Deep Compression — 25x model compression.

Global Basic Research Laboratory (NRF Project) — Participating Researcher · 2022
Deep learning-based channel estimation (CNN denoising, self-supervised learning) for IRS-aided communication systems.

ML-based Sensor Data Analysis (JS System Project) — Lead Developer · 2022
ML-based sensor feature-importance extraction and model performance optimization.

AWARDS

Best Paper Award · KNU-EE Research Congress — 2025
A Novel VLM-Guided Diffusion Model for Remote Sensing Image Super-Resolution

SKILLS

Research	model quantization, KV-cache compression, diffusion/LLM inference acceleration, split computing, super-resolution, long-context evaluation
Programming	Python, C++, CUDA C, MATLAB
Frameworks & Tools	PyTorch, HuggingFace Transformers, TensorRT, Triton, flash-attn, Docker, Git, LaTeX
Methodology	preregistration, claims ledger, freeze discipline
Languages	Korean (native) · English (business fluent)